

Problem 1

KK 1.1 - Vector Algebra 1

Given $\vec{A} = 2\hat{i} - 3\hat{j} + 7\hat{k}$ and $\vec{B} = 5\hat{i} + \hat{j} + 2\hat{k}$, we find

$$\vec{A} + \vec{B} = \begin{pmatrix} 7 \\ -2 \\ 9 \end{pmatrix} \quad (1.1)$$

$$\vec{A} - \vec{B} = \begin{pmatrix} -3 \\ -4 \\ 5 \end{pmatrix} \quad (1.2)$$

$$\vec{A} \cdot \vec{B} = 2 \cdot 5 - 3 \cdot 1 + 7 \cdot 2 = 21 \quad (1.3)$$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -3 & 7 \\ 5 & 1 & 2 \end{vmatrix} = \begin{pmatrix} -6 - 7 \\ 35 - 4 \\ 2 + 15 \end{pmatrix} = \begin{pmatrix} -13 \\ 31 \\ 17 \end{pmatrix} \quad (1.4)$$

KK 1.2 - Vector Algebra 2

Given $\vec{A} = 3\hat{i} - 2\hat{j} + 5\hat{k}$ and $\vec{B} = 6\hat{i} - 7\hat{j} + 4\hat{k}$, we find

$$|\vec{A}|^2 = 9 + 4 + 25 = 38 \quad (1.5)$$

$$|\vec{B}|^2 = 36 + 49 + 16 = 101 \quad (1.6)$$

$$(\vec{A} \cdot \vec{B})^2 = (18 + 14 + 20)^2 = 52^2 = 2704 \quad (1.7)$$

KK 1.3 - Cosine and Sine

Given $\vec{A} = 3\hat{i} + \hat{j} + \hat{k}$ and $\vec{B} = -2\hat{i} + \hat{j} + \hat{k}$, we find

$$\cos \theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}||\vec{B}|} = \frac{-6 + 1 + 1}{\sqrt{9 + 1 + 1}\sqrt{4 + 1 + 1}} = \frac{-4}{\sqrt{11} \cdot \sqrt{6}} = -\frac{4}{\sqrt{66}} \quad (1.8)$$

$$\sin \theta = \frac{|\vec{A} \times \vec{B}|}{|\vec{A}||\vec{B}|} = \frac{1}{\sqrt{66}} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & 1 \\ -2 & 1 & 1 \end{vmatrix} = \frac{1}{\sqrt{66}} \left| \begin{pmatrix} 0 \\ -5 \\ 5 \end{pmatrix} \right| = \frac{5}{\sqrt{33}} \quad (1.9)$$

An alternative creative solution is to use the trig identity $\sin^2 \theta + \cos^2 \theta = 1$ to solve for sine from the dot product

$$\sin \theta = \sqrt{1 - \frac{16}{66}} = \sqrt{\frac{50}{66}} = \frac{5}{\sqrt{33}} \quad (1.10)$$

Problem 2

KK 1.9 - Perpendicular Unit Vector

Consider $\vec{A} = \hat{i} + \hat{j} - \hat{k}$ and $\vec{B} = 2\hat{i} + \hat{j} - 3\hat{k}$. We wish to find a unit vector perpendicular to both. It suffices to take a cross product and normalize

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -1 \\ 2 & 1 & -3 \end{vmatrix} = \begin{pmatrix} -3 + 1 \\ -2 + 3 \\ 1 - 2 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \\ -1 \end{pmatrix} \quad (2.1)$$

Therefore

$$\hat{n} = \frac{1}{\sqrt{6}} \begin{pmatrix} -2 \\ 1 \\ -1 \end{pmatrix} \quad (2.2)$$

Note there are two such unit vectors, and $-\hat{n}$ is also perpendicular to $\vec{A}$ and $\vec{B}$.

KK 1.11 - Parallelepiped

A parallelepiped is a 3D parallelogon. In general, there are 3 distinct side lengths. Its volume is equal to the area of one face times the height of the orthogonal projection.

Let us pick any corner and define vectors $\vec{A}$, $\vec{B}$, and $\vec{C}$ to point along the three side lengths from that corner. The area of one face is the magnitude $|\vec{A} \times \vec{B}|$. Its height is the projection $\vec{C} \cdot \hat{n}$, where $\hat{n}$ is the unit vector along $\vec{A} \times \vec{B}$. Therefore the volume is $(\vec{A} \times \vec{B}) \cdot \vec{C}$.

Problem 3

KK 1.17 - Rolling Drum

The physical constraint “rolling without slipping” means $\Delta s = R\Delta\theta$. First, we ask the question - does the angle of the slope matter? No, it does not, because the acceleration a is already defined. If we had to solve for a using Newton’s law, we may have found $a = g \sin \theta$.

Next we ask the question, how does angular acceleration relate to the center of mass motion? This follows immediately from the no-slip condition. Each full 2π revolution results in $\Delta s = 2\pi R$ translation.

Therefore, in a coordinate frame angled parallel to the slope, $x(t) = R\theta(t)$. Consequently, $a(t) = R\alpha$. So we conclude

$$\alpha = a/R \quad (3.1)$$

Problem 4

KK 1.23 - Smooth Elevator

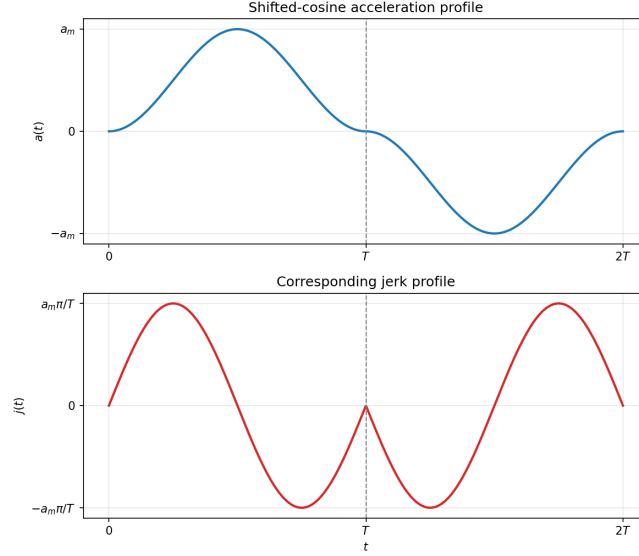
a) sketch

This problem investigates the jerk, which is the 3rd time derivative of position. We are given

$$\begin{aligned} a(t) &= \frac{a_m}{2} [1 - \cos(2\pi t/T)] & 0 \leq t \leq T \\ a(t) &= -\frac{a_m}{2} [1 - \cos(2\pi t/T)] & T \leq t \leq 2T \end{aligned} \quad (4.1)$$

This strange acceleration takes the form of a shifted cosine. If $j(t) = \dot{a} = \ddot{x}$, then

$$\begin{aligned} j(t) &= \frac{a_m \pi}{T} \sin(2\pi t/T) & 0 \leq t \leq T \\ j(t) &= -\frac{a_m \pi}{T} \sin(2\pi t/T) & T \leq t \leq 2T \end{aligned} \quad (4.2)$$



How does one draw this with only pen and paper? We see that $(1 - \cos x)$ looks like an upside-down cosine, shifted up such that the full period is positive. Then the second period of $a(t)$ must be entirely negative. This justifies the odd piece-wise definition, there is a discontinuity in the derivative at $t = T$.

For $j(t) = \dot{a}$ that discontinuity becomes apparent. The vertical offset drops after the derivative, and we have two sines mirrored horizontally about $t = T$.

b) max speed

Speed is the magnitude of velocity. Since this is a 1D problem, it is just the absolute value. We integrate

$$\begin{aligned} v(t) &= \frac{a_m}{2} \left[t - \frac{T}{2\pi} \sin(2\pi t/T) \right] & 0 \leq t \leq T \\ v(t) &= \frac{a_m}{2} \left[2T - t + \frac{T}{2\pi} \sin(2\pi t/T) \right] & T \leq t \leq 2T \end{aligned} \quad (4.3)$$

The max speed is achieved at $t = T$. We can check this visually, by confirming the area under the curve is maximal just before $a(t)$ goes negative. Since $\sin 2\pi = 0$, we find

$$v_{max} = v(T) = \frac{a_m T}{2} \quad (4.4)$$

c) initial speed

To find the initial speed when $t \ll T$ we approximate

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \quad (4.5)$$

This is an asymptotic expansion. We can drop the higher order terms when $x \rightarrow 0$, which is the case for $x = 2\pi t/T$ at early times. The first term cancels, so we keep the second order correction.

$$t - \frac{T}{2\pi} \sin(2\pi t/T) = t - \left[t - \frac{T}{2\pi} \frac{1}{3} \left(\frac{2\pi t}{T} \right)^3 + \dots \right] \approx \frac{2\pi^2}{3T^2} t^3 \quad (4.6)$$

Therefore

$$v(t \rightarrow 0) = \frac{\pi^2}{3T^2} a_m t^3 \quad (4.7)$$

d) total time

We integrate again to find position. Since the velocity profile is symmetric about $t = T$, it suffices to solve only half the integral

$$\begin{aligned} D = \Delta x = x(2T) &= 2 \int_0^T v(t) dt \\ &= a_m \left[\frac{t^2}{2} + \left(\frac{T}{2\pi} \right)^2 \cos(2\pi t/T) \right]_0^T \\ &= \frac{1}{2} a_m T^2 \end{aligned} \quad (4.8)$$

Therefore $T = \sqrt{2D/a_m}$.

Problem 5 (Bonus)

KK 1.26 - Rock on a Hill

We wish to find optimal launch angle $\theta(\phi)$, given a slope of angle ϕ . How can we make this problem easier? When $\phi = 0$, we are doing a familiar projectile motion problem.

simple case ($\phi = 0$)

Let us consider

$$\vec{x}(t) = v_0 \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix} t - \frac{1}{2} g \begin{pmatrix} 0 \\ 1 \end{pmatrix} t^2 = \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix} \quad (5.1)$$

For $\phi = 0$, $\Delta y = 0$ because the ground is flat. So we can constrain the flight time

$$\Delta y = v_0 T \sin \theta - \frac{1}{2} g T^2 = 0 \quad (5.2)$$

and derive

$$T_{\phi=0} = \frac{2v_0}{g} \sin \theta \quad (5.3)$$

It follows then that the distance is

$$\begin{aligned} \Delta x &= v_0 T \cos \theta \\ &= 2 \frac{v_0^2}{g} \cos \theta \sin \theta \\ D &= \frac{v_0^2}{g} \sin 2\theta \end{aligned} \quad (5.4)$$

At this point one could take a derivative $dD/d\theta$ to optimize the distance. Or one could recognize that $D(\theta)$ is maximal when $\sin 2\theta = 1$. Therefore $\theta = \pi/4$, indicating 45 degree launch.

general case ($\phi > 0$)

Now that we've gained some intuition, we can tackle the general case where $\Delta y < 0$. We have a new constraint

$$\Delta y = v_0 T \sin \theta - \frac{1}{2} g T^2 = -D \tan \phi \quad (5.5)$$

$$\Delta x = v_0 T \cos \theta = D \quad (5.6)$$

We could solve Δy as a quadratic equation for T , but it is actually easier to just substitute $T = \frac{D}{v_0 \cos \theta}$ from the Δx equation. Then

$$D \tan \theta - \frac{g D^2}{2 v_0^2 \cos^2 \theta} = -D \tan \phi \quad (5.7)$$

and we can divide through by D and rearrange the equation to find

$$\begin{aligned} \left(\frac{g}{2 v_0^2 \cos^2 \theta} \right) D &= \tan \theta + \tan \phi \\ \frac{g}{v_0^2} D &= 2 \cos^2 \theta (\tan \theta + \tan \phi) \\ \bar{D} &= \sin 2\theta + 2 \cos^2 \theta \tan \phi \end{aligned} \quad (5.8)$$

Note how the LHS motivates the definition of a dimensionless distance $\bar{D} = gD/v_0^2$. We now take a derivative to optimize

$$\begin{aligned} 0 &= \frac{\partial \bar{D}}{\partial \theta} = 2 \cos 2\theta - 4 \sin \theta \cos \theta \tan \phi \\ &= 2 [\cos 2\theta - \sin 2\theta \tan \phi] \end{aligned} \quad (5.9)$$

Finally we find

$$\tan 2\theta = \cot \phi \quad (5.10)$$

Therefore the specific relation $\theta(\phi)$ is

$$\theta = \frac{1}{2} \tan^{-1} (\cot \phi) \quad (5.11)$$

We may confirm that $\phi = 0$ implies $\cot \phi = \infty$, such that $\tan^{-1} = \pi/2$ and $\theta = \pi/4$.

A more elegant solution can be found by recognizing $\cot \phi = \tan(\frac{\pi}{2} - \phi)$. This “shift and flip” allows us to cancel both tangents and find

$$\theta = \frac{\pi}{4} - \frac{\phi}{2} \quad (5.12)$$